

King Saud University

College of Computer and Information Sciences

Computer Engineering Department

CEN415: Introduction to VLSI Design

8-bit input ALU

by

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1 Introduction

A combinational circuit which makes a specific arithmetic operations.

2 Problem Statement and Specifications

It's required to design a system which is capable of doing the operations shown in table below:

Operation code		Operation (Y)	Description
s1	s0		
0	0	A * B	Multiply A by B using parallel CSA and CPA
0	1	A * B	Multiply A by B using shift and add.
1	0	A - B	Simple subtraction
1	1	5 * (A+B)	Multiply the addition result by 5

The system must have (at least): two different 8-bit inputs, 2-bit selectors, Common V_{DD} & GND, 16-bit output, Carry out signal.

Table 1. Operations of ALU

3 Motivation

The motivation is to apply most of the CMOS VLSI design fundamental concepts, such as using a layout tool to make layout of real circuit and gates.

4 Digital Logic Design

This section shows the digital logic design of the 8-bit system using Logisim, which is a tool for designing and simulating logic circuits.

4.1 Complete Design

Figure below shows the complete design of the circuit.

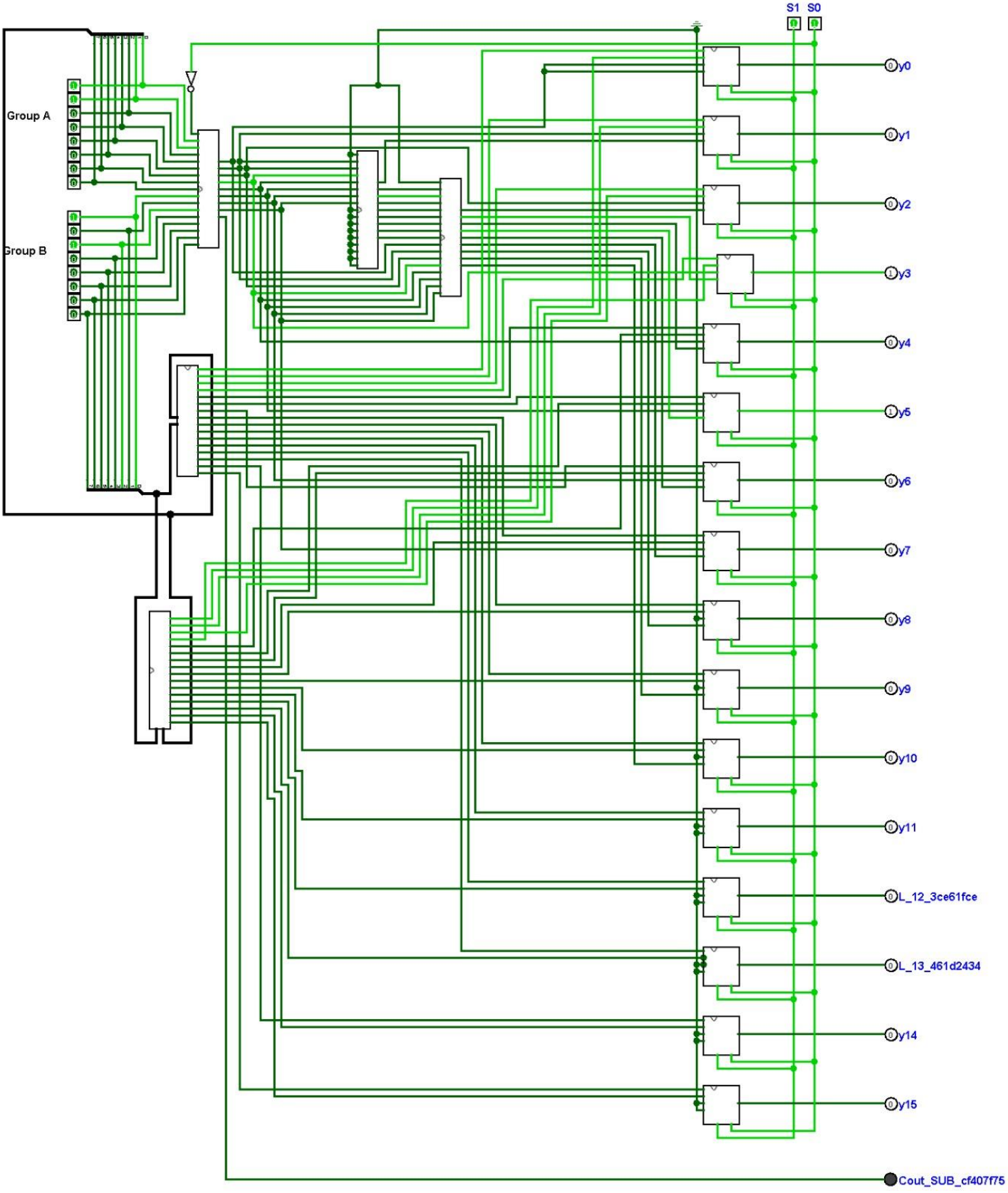


Figure 1. Complete Design

4.2 Multiply A by B using parallel CSA and CPA Unit

Figures below show the design of the Multiply A by B using parallel CSA and CPA unit.

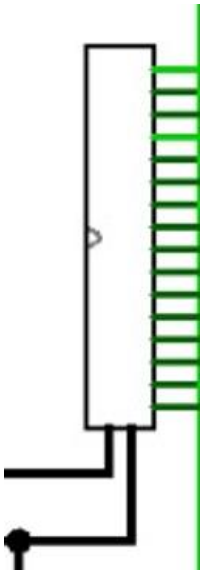


Figure 2. A*B Unit (Lvl 0)

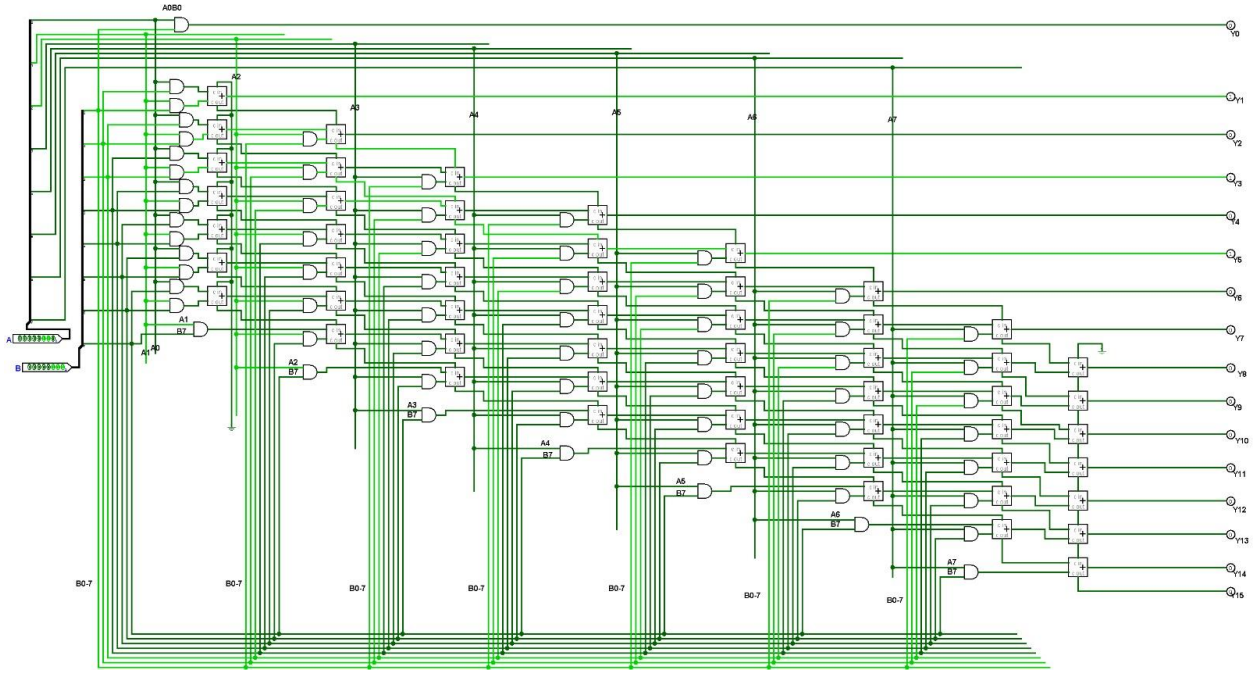


Figure 3. A*B Unit (Lvl 1)

4.3 Multiply A by B using shift and add.



Figure 5. A*B (Lvl 0)

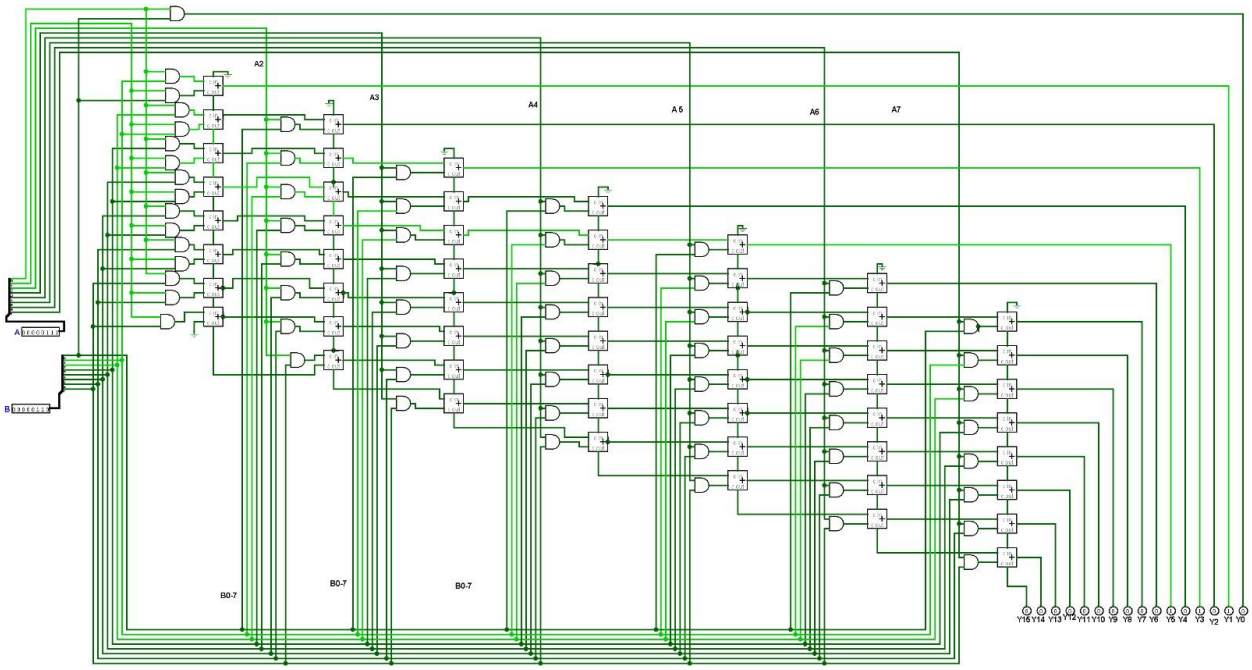


Figure 6. A*B (Lvl1)

4.4 Subtractor

In logic design we know that A – B = A + (~B) + 1, this technique is used in this subtractor by inverting the inputs of B and then adding them to A, which can be accomplished by inverting the S0 to choose between subtracting or adding, figures below show the subtraction unit.

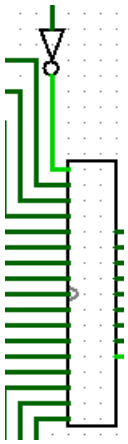


Figure 8. Subtraction Unit (Lvl 0)

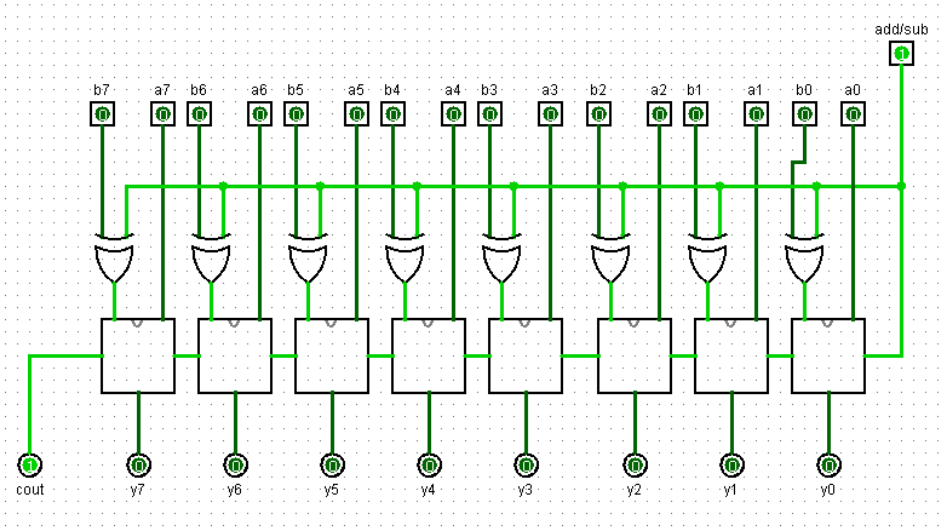


Figure 9. Subtraction Unit (Lvl 1)

4.5 Multiply the addition result by 5

The adder used is of type (Carry Ripple Adder), which is a two 8-bit inputs adder composed of 8 full adders having each carry out signal connected to the next adders as a carry in, then multiply it by constant number 5 by using two full adders and hardly connecting the multiplication. figures below present the design of the adder used.

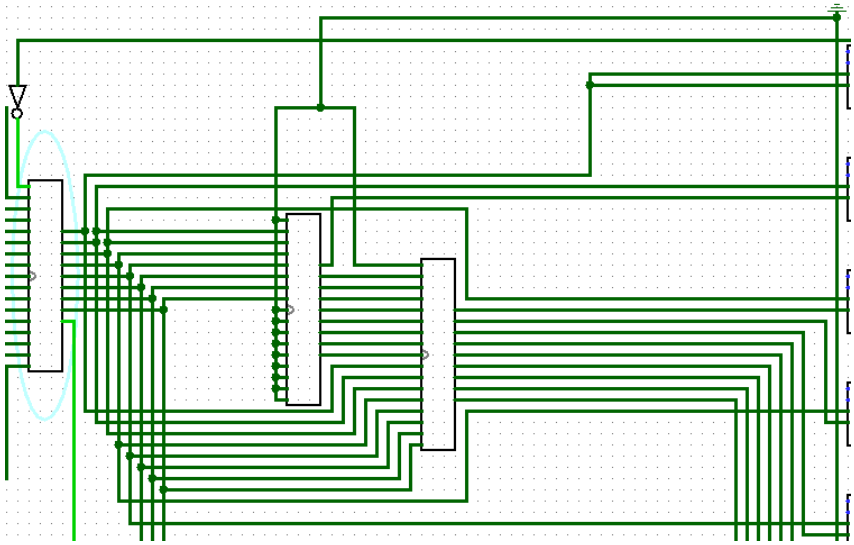


Figure 10. Multiply the addition result by 5 Unit (Lvl 0)

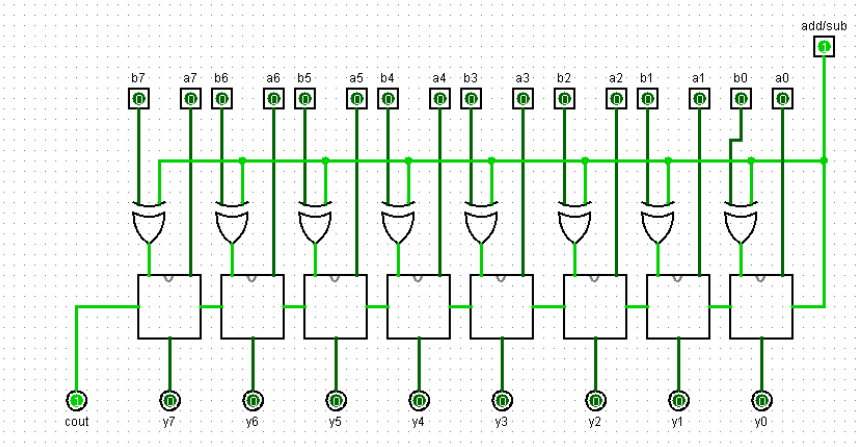


Figure 11. FULL adder (Lvl 1)

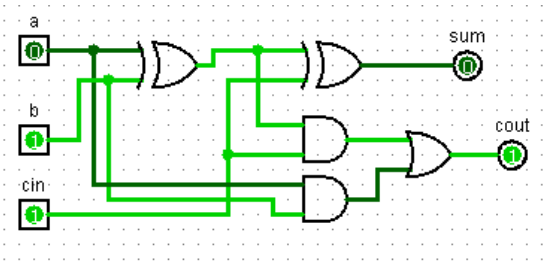


Figure 11. FULL adder (Lvl 2)

4.6 Multiplexer

The final design component of this section is the multiplexer, the multiplexer used Is a 4:1 multiplexer made out of 3 smaller 2:1 multiplexers, figures below present both levels of design for this unit.

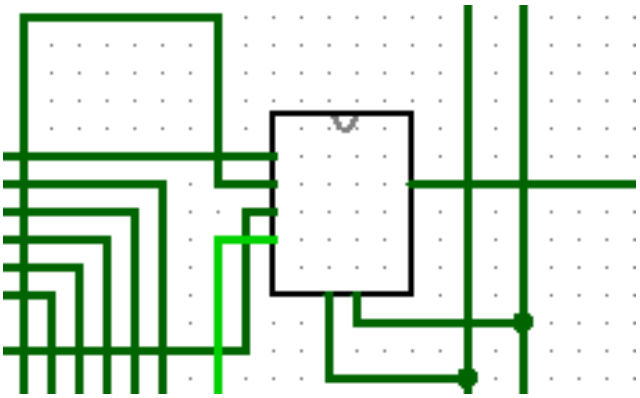


Figure 12. Multiplexer (Lvl 0)

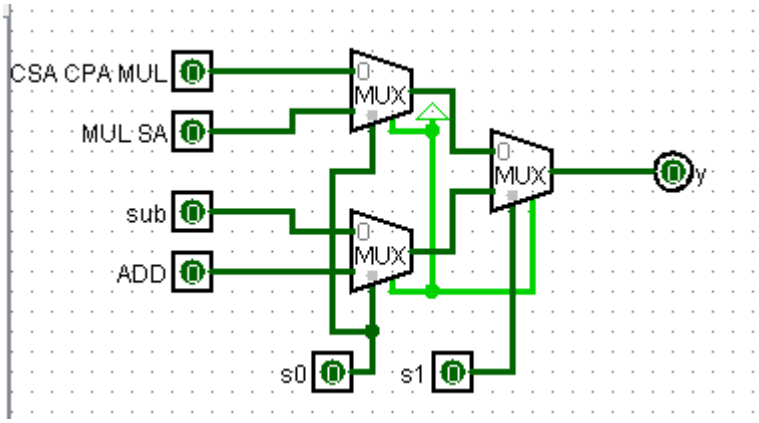
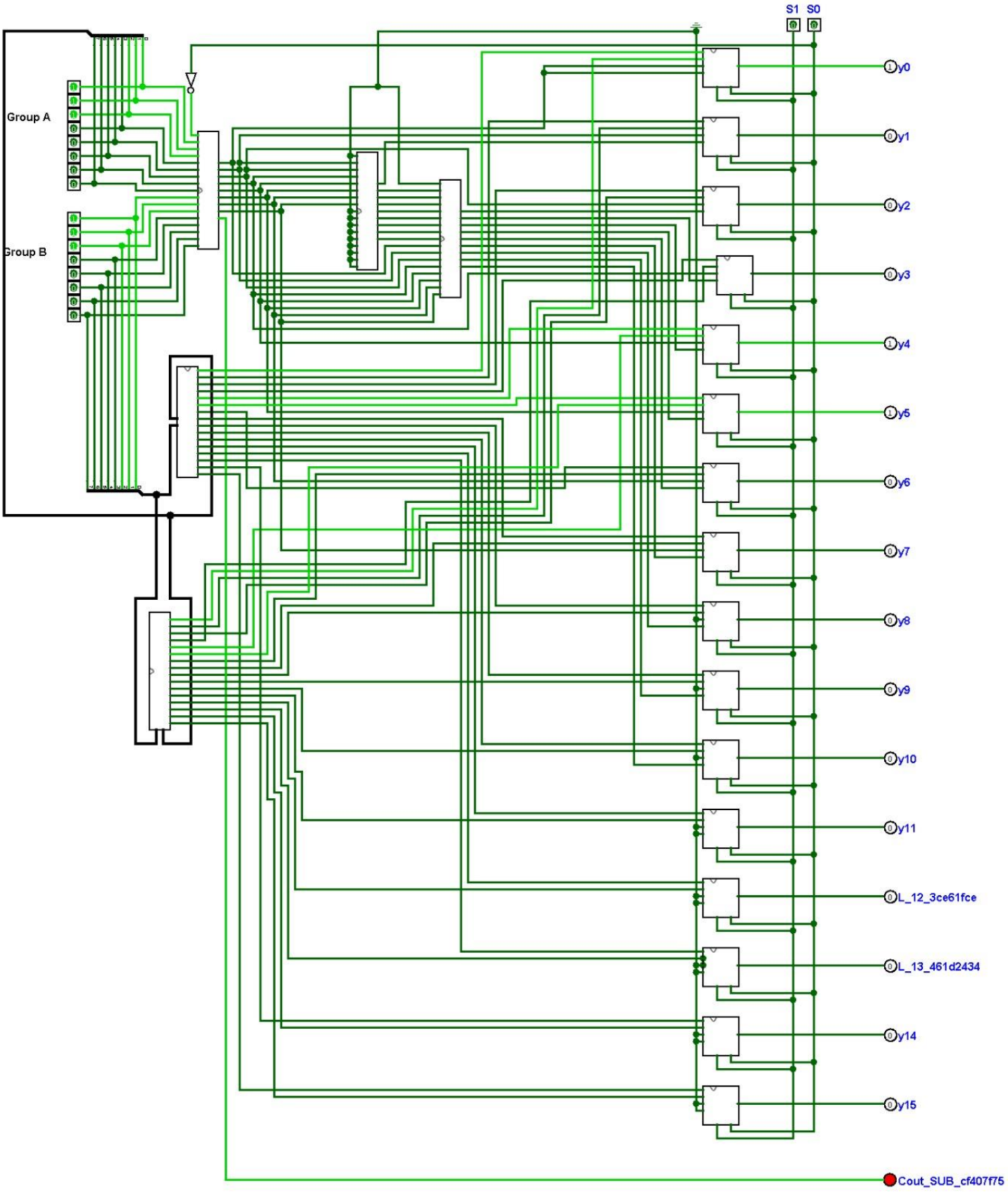


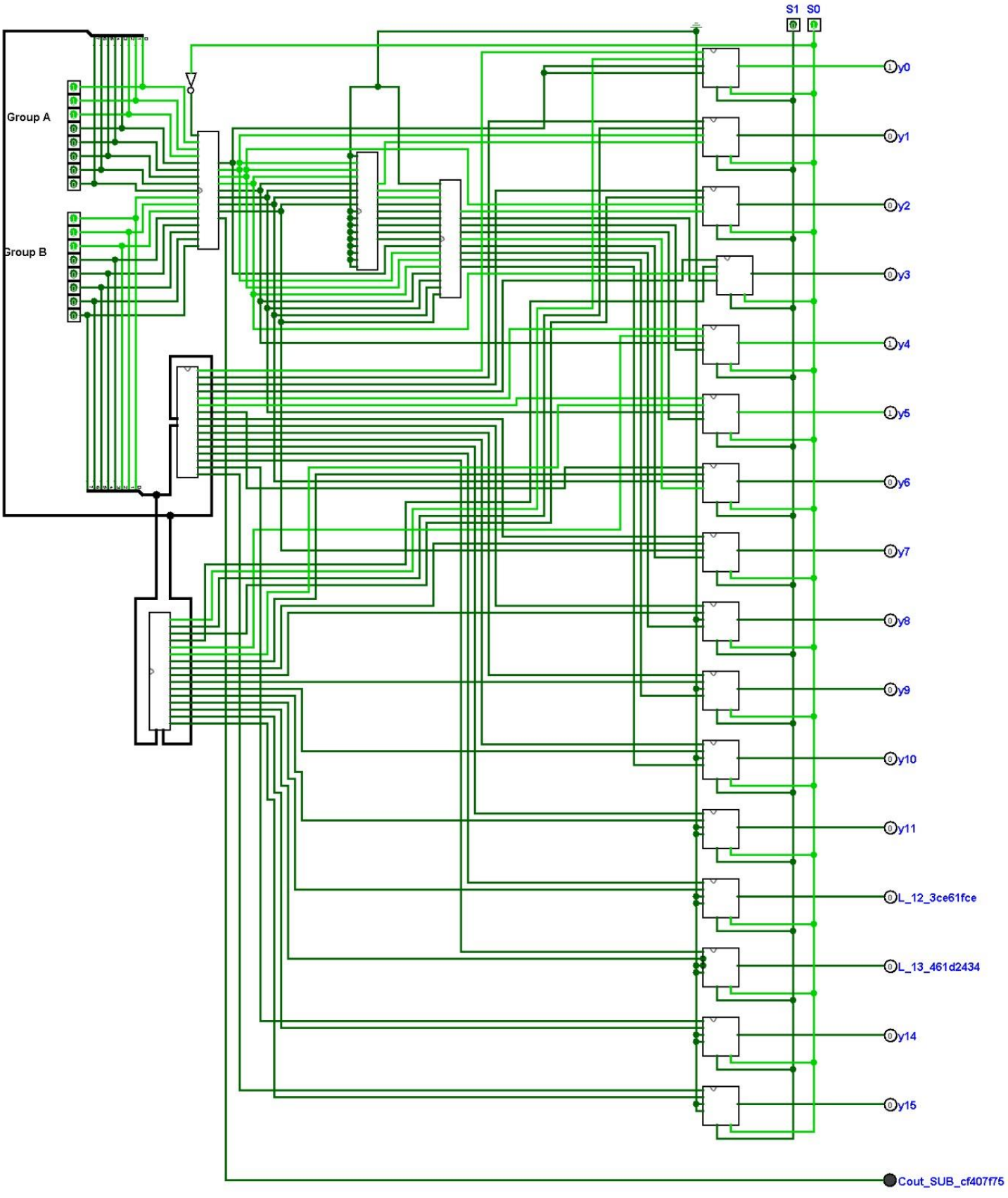
Figure 13. Multiplexer (Lvl 1)

Sample inputs verification

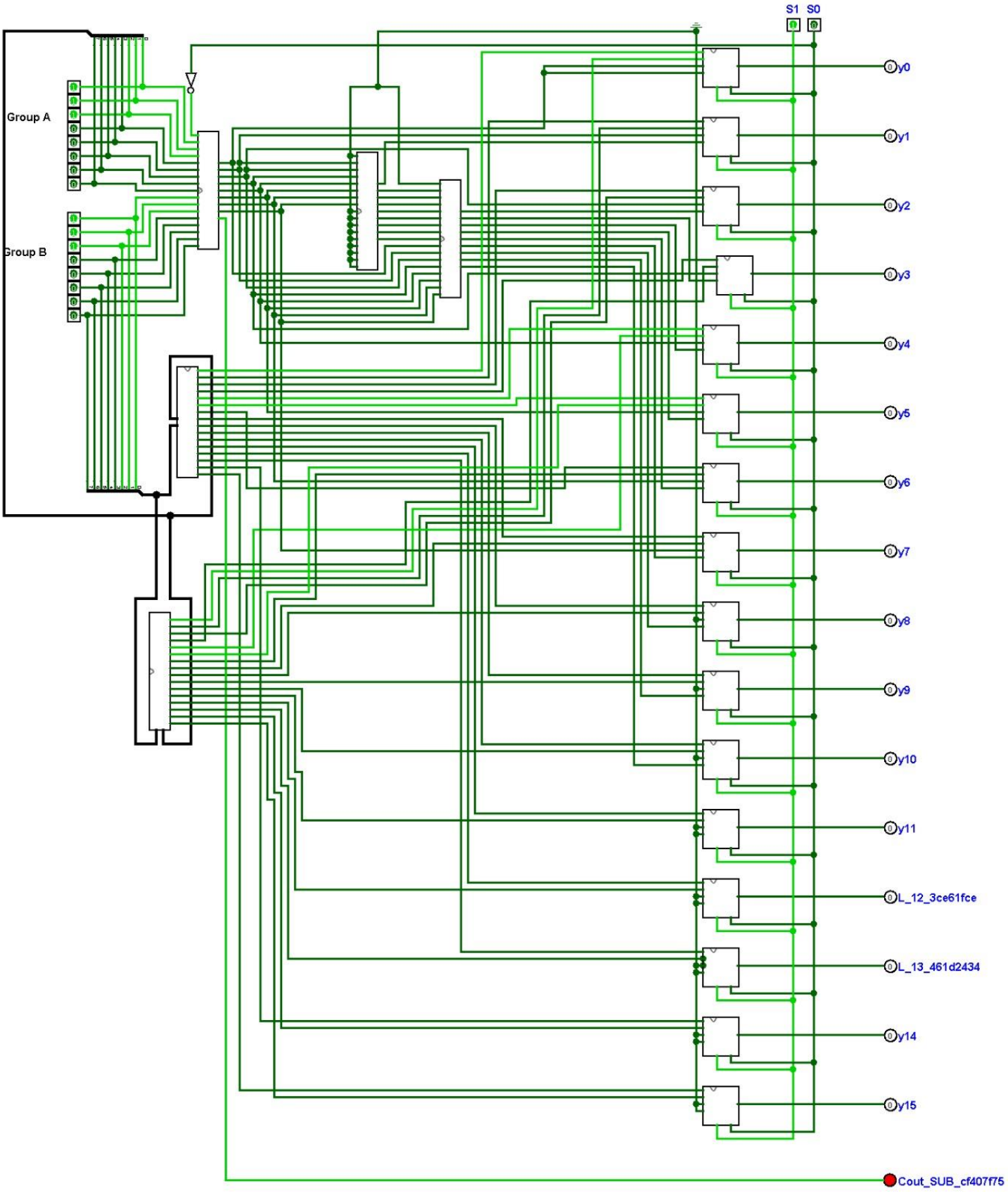
7*7 using CSA CPA



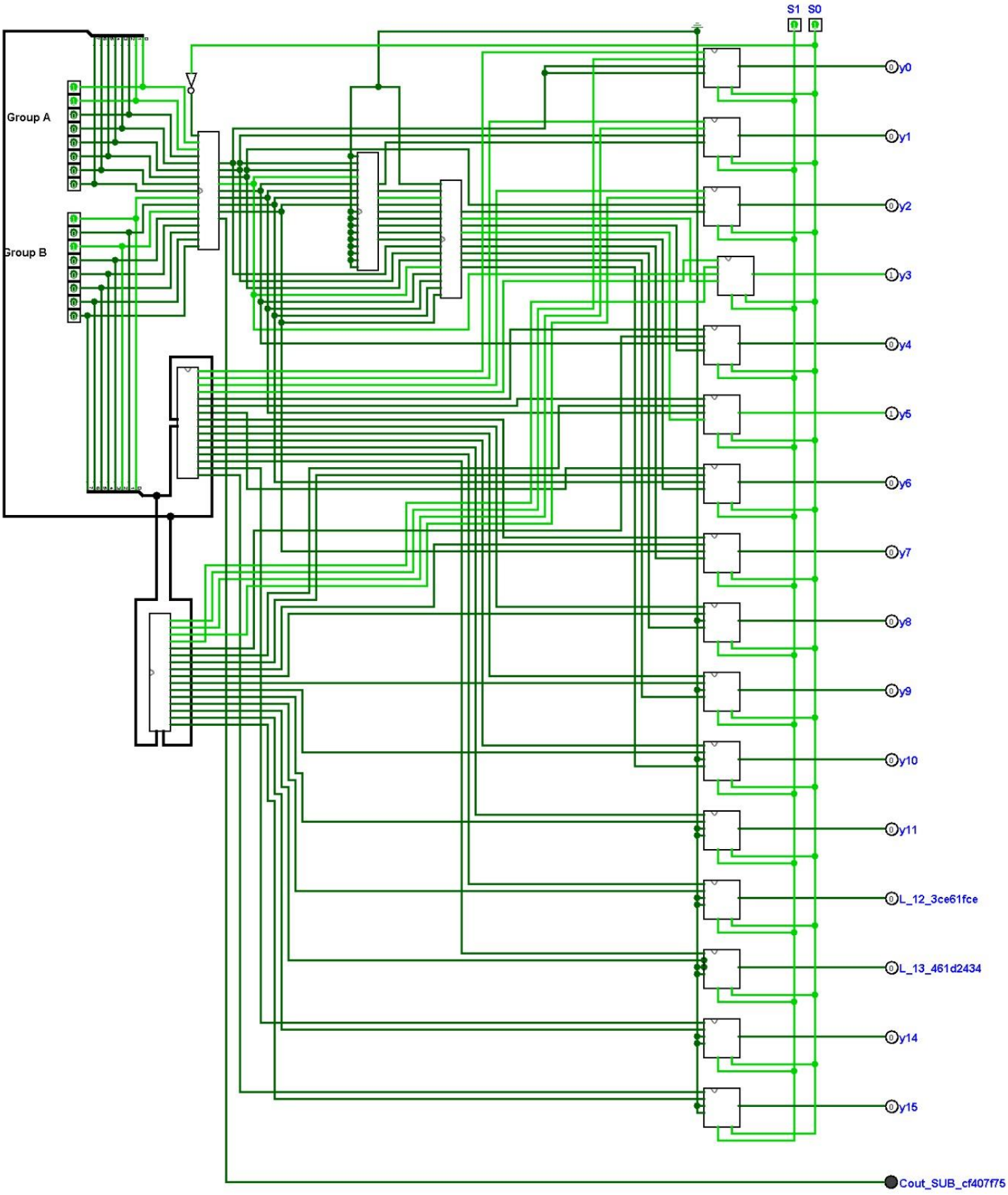
7*7 using shiftandadd



Sub 7-7

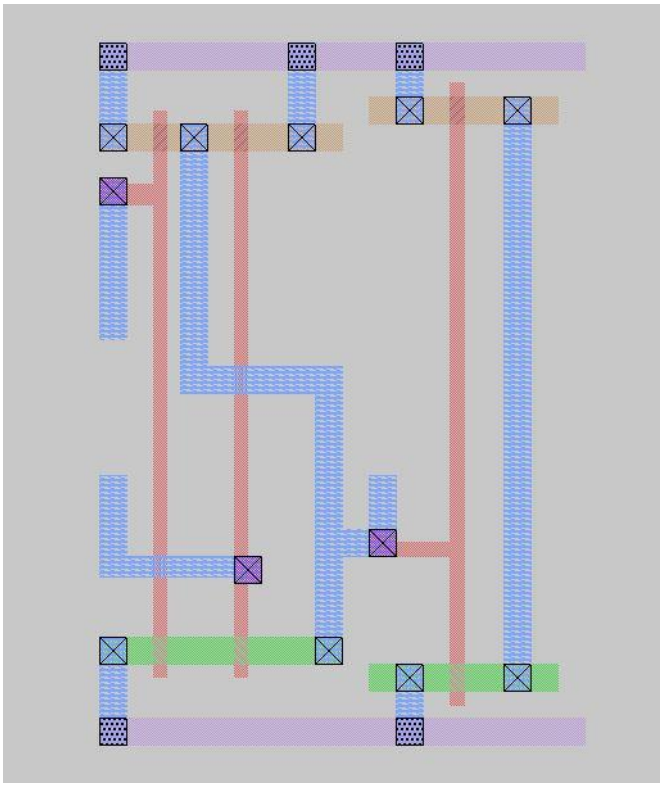

S1
S0
Group A
Group B
y0
y1
y2
y3
y4
y5
y6
y7
y8
y9
y10
y11
L_12_3ce61fce
L_13_461d2434
y14
y15
Cout_SUB_cf407f75

Add 3+5 and multiply by 5

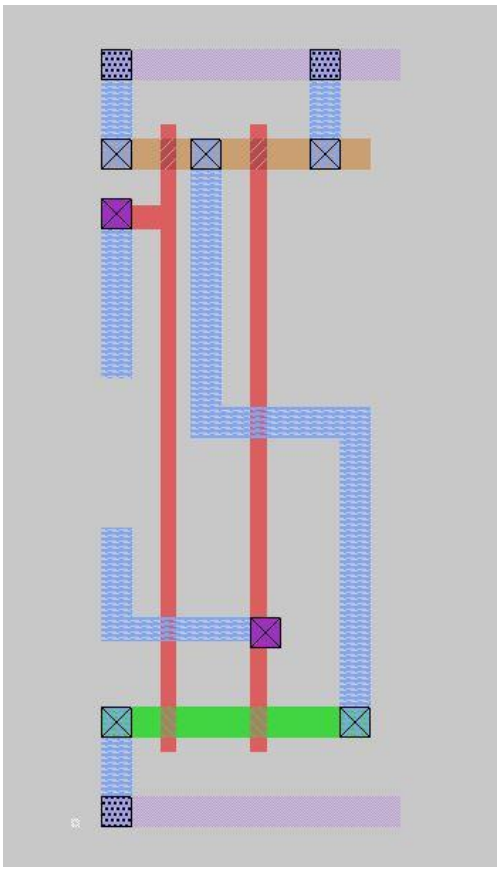


5. Stick diagram:

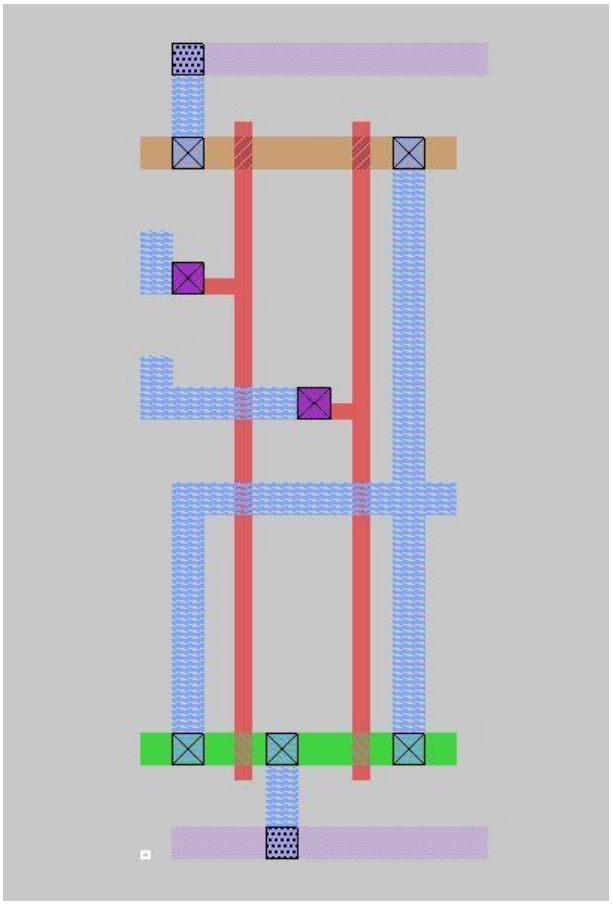
AND:



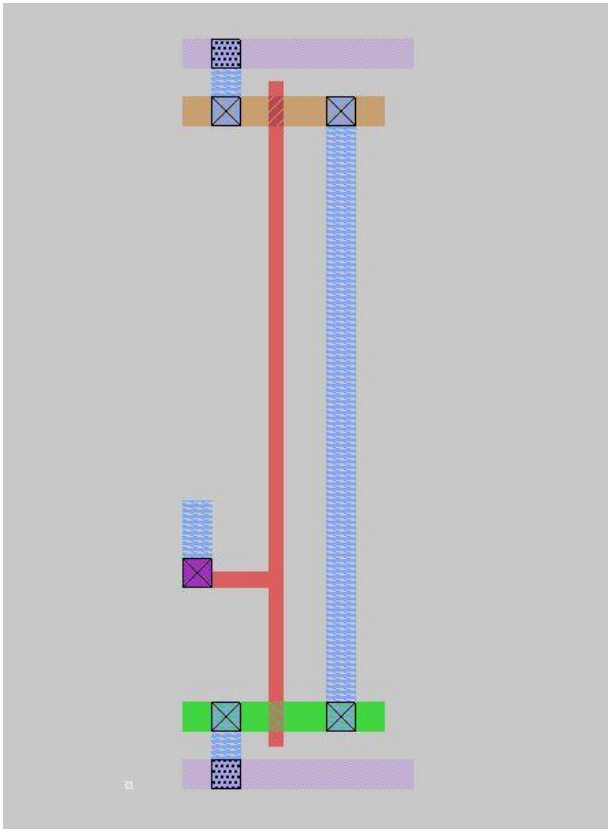
NAND:



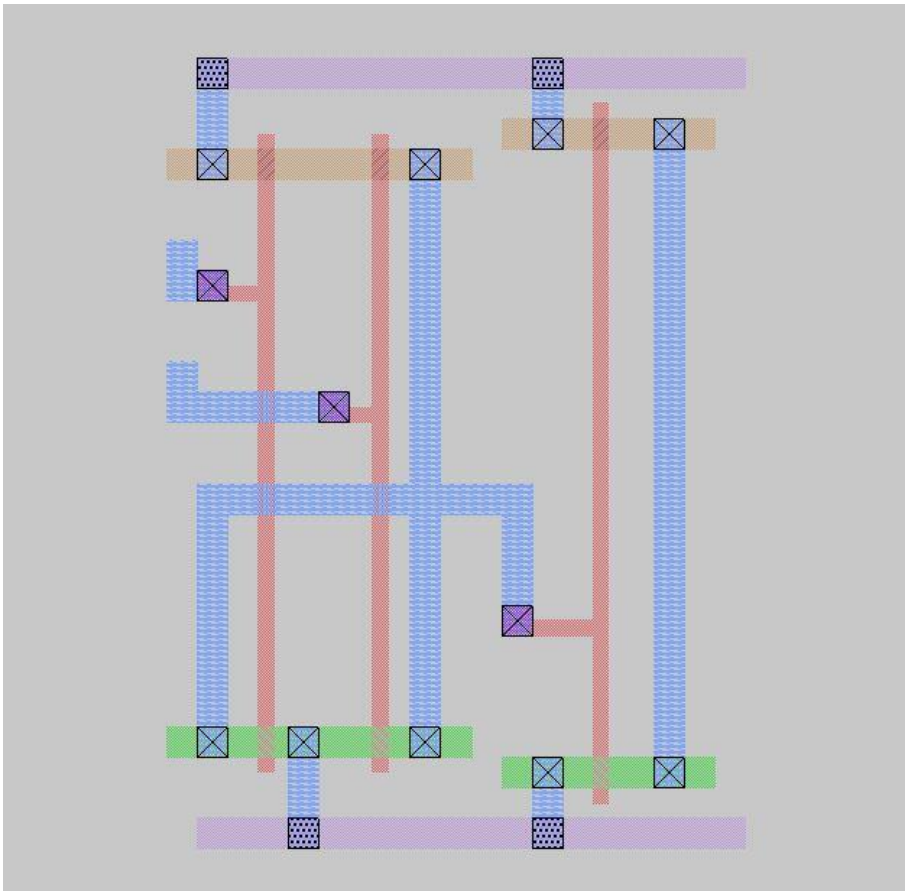
NOR:



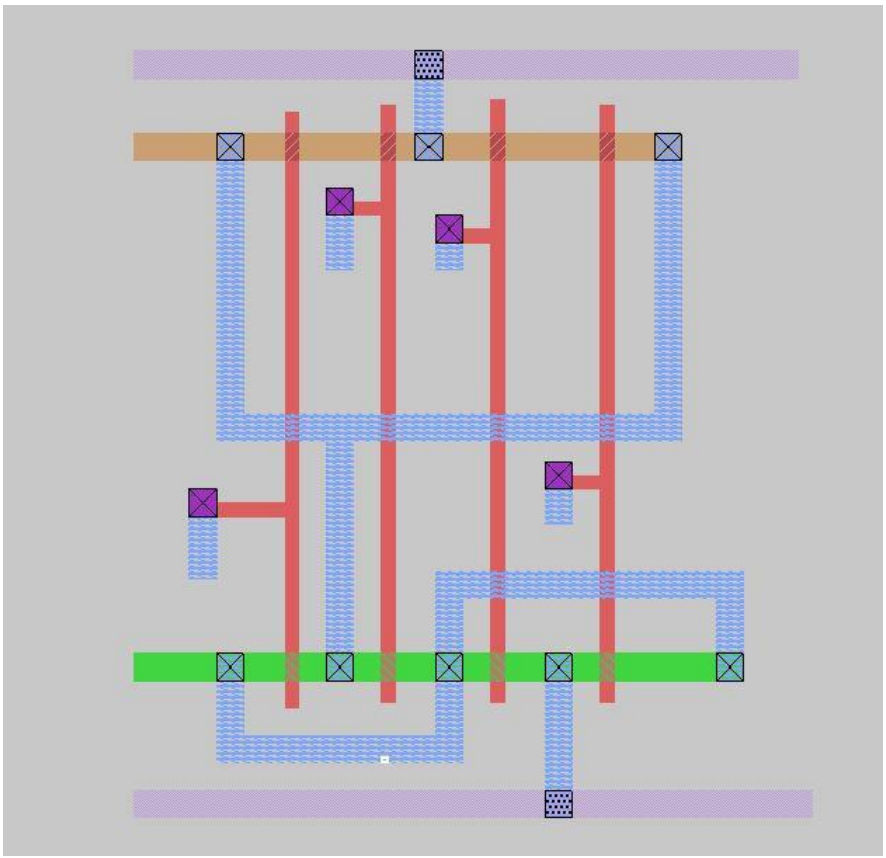
NOT:



OR:



XOR:



6. Testing

System Consist of :

16 4-1 Mux

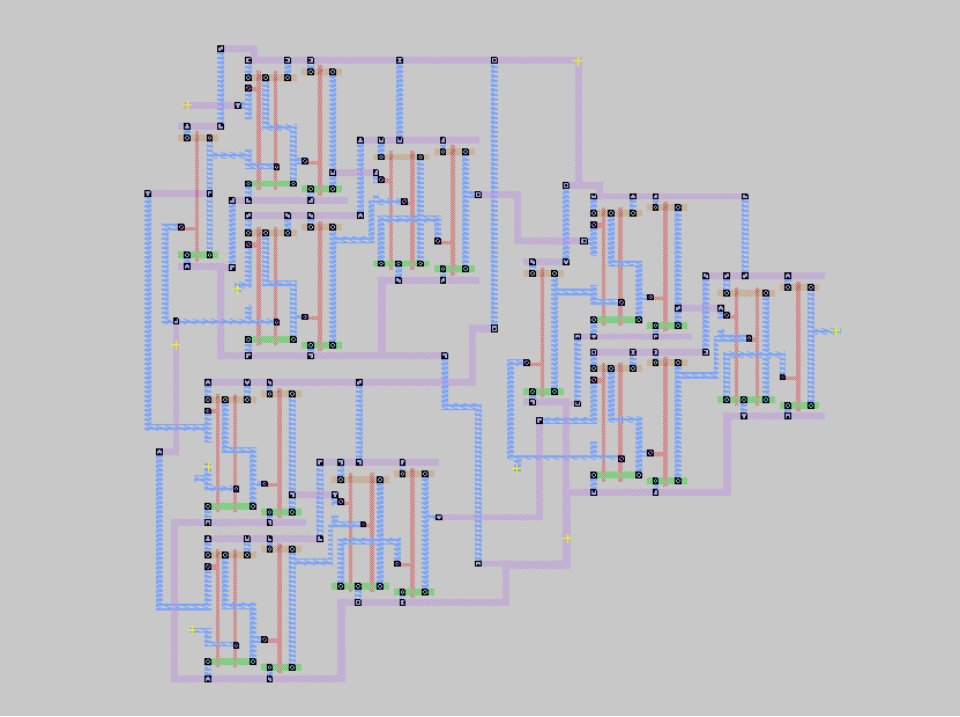
8bitSubsracter

8bitAdder-Multiplierb5

8bitCsaCpaMultiplier

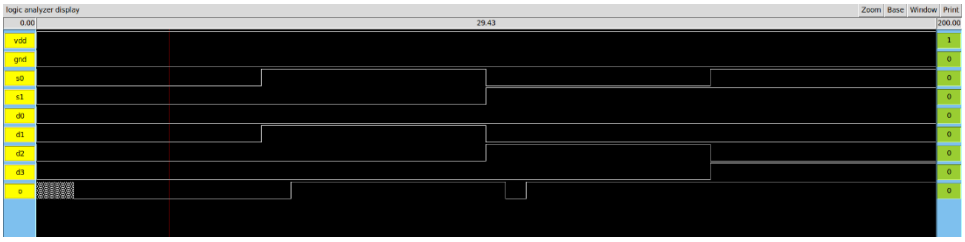
8bitShiftAndAddMultiplier

4-1 Mux :

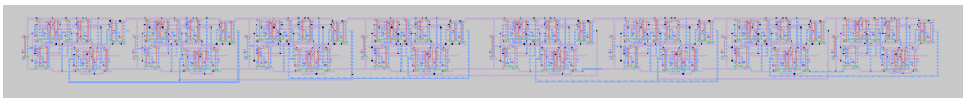


Test :

```
stepsize 50
ana vdd gnd s0 s1 d0 d1 d2 d3 o
h vdd
l gnd
l s0
l s1
l d0
l d1
l d2
l d3
s
h s0
l s1
l d0
h d1
l d2
l d3
s
l s0
h s1
l d0
l d1
h d2
l d3
s
h s0
h s1
l d0
l d1
l d2
h d3
s
```

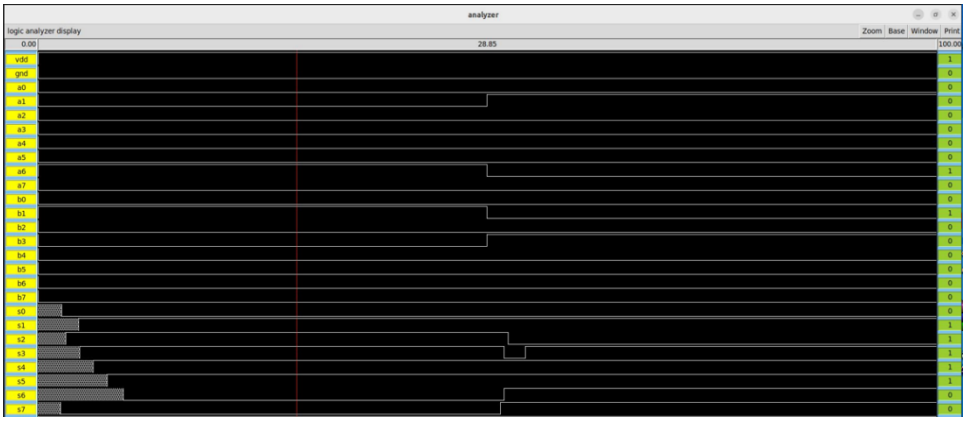


8bitSubsracter :

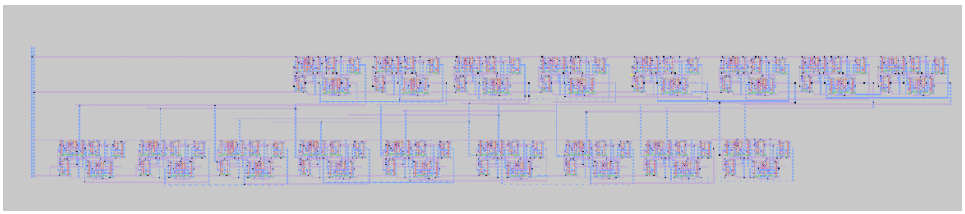


Test :

```
stepsize 50
ana vdd gnd a0 a1 a2 a3 a4 a5 a6 a7 b0 b1 b2 b3 b4 b5 b6 b7 s0 s1 s2 s3 s4 s5 s6 s7
h vdd
l gnd
l a0
l a1
l a2
l a3
l a4
l a5
h a6
l a7
l b0
h b1
l b2
l b3
l b4
l b5
l b6
l b7
s
l a0
h a1
l a2
l a3
l a4
l a5
l a6
l a7
l b0
l b1
l b2
h b3
l b4
l b5
l b6
l b7
s
```

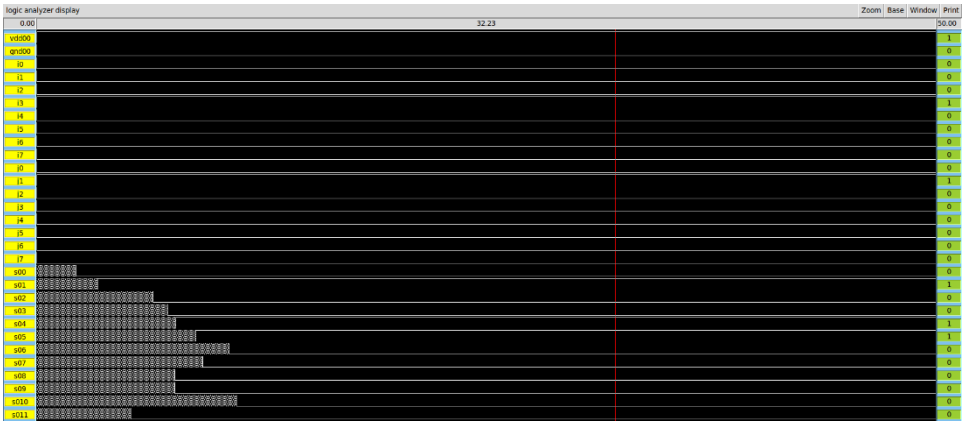


8bitAdderMultiplier5 :

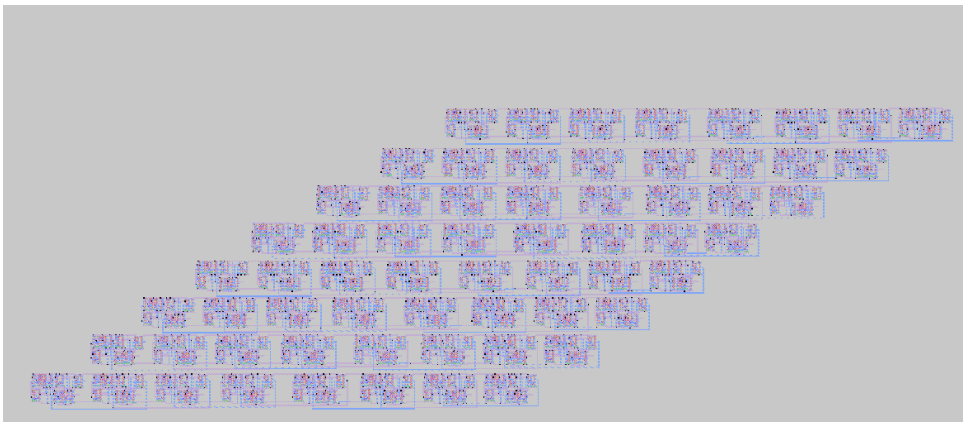


Test :

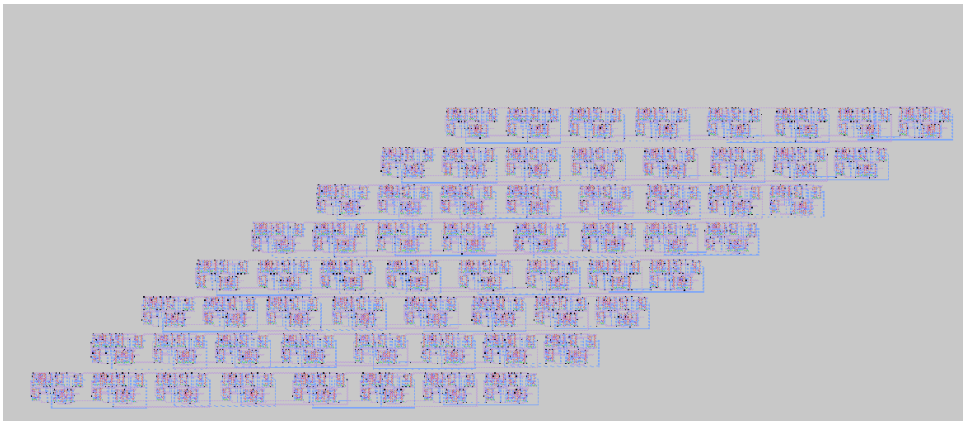
```
1 stepsize 50
2 ana vdd00 gnd00 i0 i1 i2 i3 i4 i5 i6 i7 j0 j1 j2 j3 j4 j5 j6 j7 s00 s01 s02 s03 s04 s05 s06 s07
  s08 s09 s010 s011
3 h vdd00
4 l gnd00
5 h i3
6 l i0 i1 i2 i4 i5 i6 i7
7 h j1
8 l j0 j2 j3 j4 j5 j6 j7
9 s
```



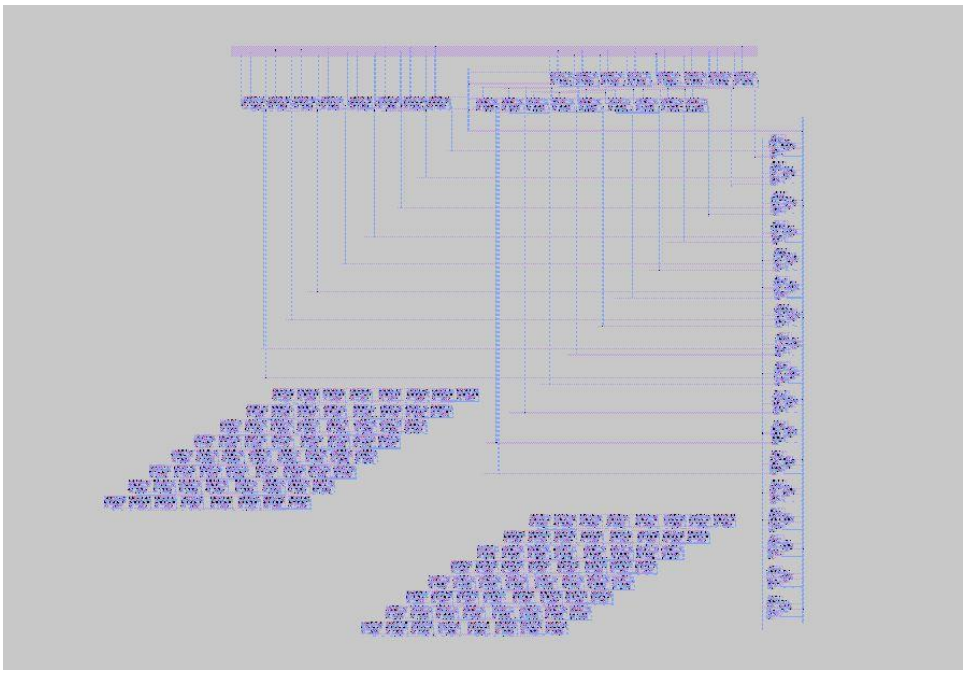
8bitCsaCpaMultiplier :



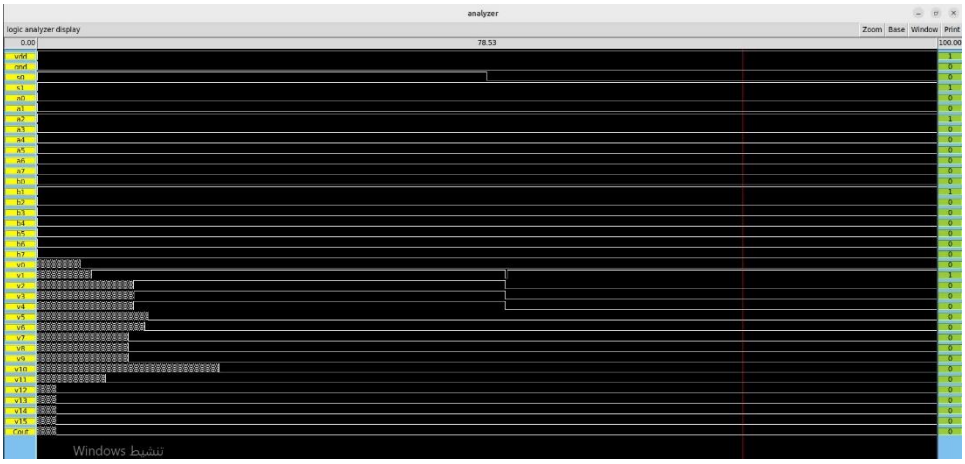
8bitShiftAndAddMultiplier :

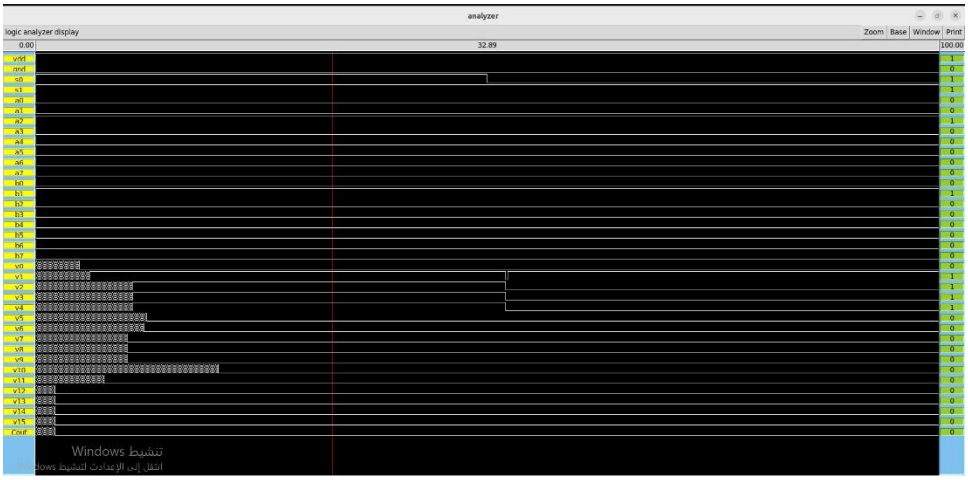


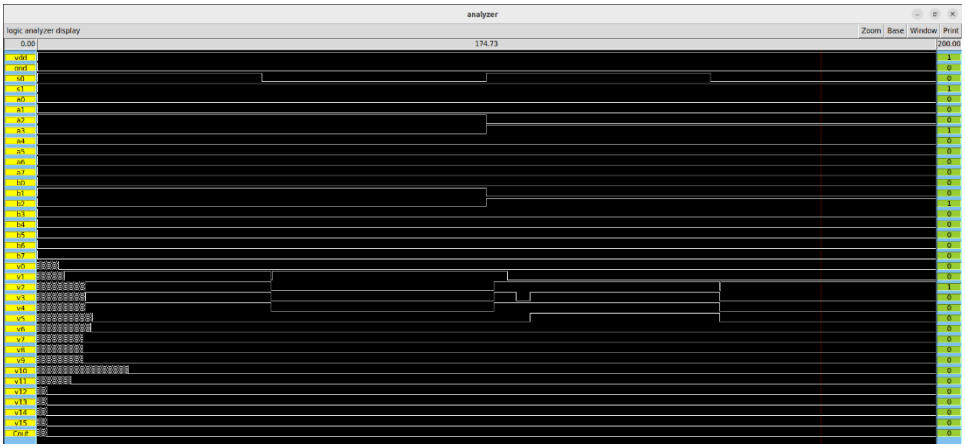
Final Design :



Test :







To see the design :

Go to the directory of the project file

Type: magic Final

To test the design :

Go to the directory of the project file

Type: irsim scmos100.prm Final.sim -Final.cmd